Multi-Pedestrian Tracking in Occluded Environments Using Vehicular Communication and Set-Based Estimation

Davide Nascivera[1], Vandana Narri[1,2], M. Umar B. Niazi[1], and Karl H. Johansson[1]

***Abstract*—A critical challenge for connected and automated vehicles is the accurate detection and robust prediction of multiple pedestrians occluded from their field of view. This paper proposes a new framework that employs Vehicle-to-Everything (V2X) communication and reliably associates and fuses measurements from multiple V2X units with varying levels of bounded uncertainty. A robust set-based estimation algorithm is developed to track, fuse, and predict the motion of multiple pedestrians. Cost-based association techniques ensure continuous tracking by linking new measurements to existing tracks and matching tracks across V2X units. The proposed framework is evaluated on a simulation platform using a public pedestrian trajectory dataset, demonstrating that it provides tight, guaranteed enclosures of pedestrian positions while outperforming standard techniques in terms of accuracy.**

***Keywords*— Collaborative Perception; Sensor Fusion; Pedestrian Safety; Set-Based Estimation.**

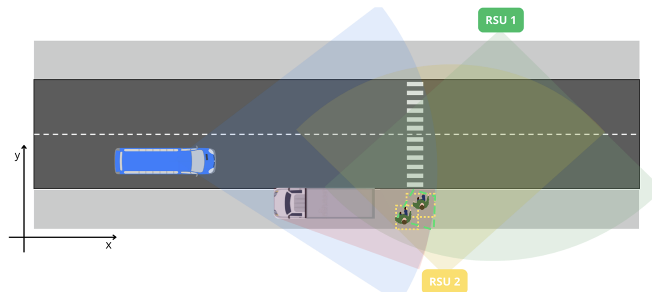

Fig. 1. A scenario with an ego-vehicle, a parked vehicle, and two RSUs. The field of view of the ego-vehicle, RSU 1, and RSU 2 are highlighted by blue, green, and yellow sectors, respectively. The red sector highlights the occluded region in the ego-vehicle's field of view. Two pedestrians who are about to cross the road are occluded by the parked vehicle. The measurements from RSU 1 and RSU 2 are represented by green and yellow polygons.

I. Introduction

As connected and automated vehicles (CAVs) advance toward level 5 autonomy, occlusions caused by parked vehicles, infrastructure, and other traffic participants limit the ego-vehicle's situational awareness and increase risks to pedestrians. Limited situational awareness can compromise the ego-vehicle's ability to plan safe trajectories [1]. V2X communication addresses this problem by enabling the sharing of perception data from multiple V2X units, which is critical in complex urban scenarios with occluded pedestrians. Previous work [2] employed set-based methods to fuse shared data to provide robust safety guarantees for a single occluded pedestrian. However, the problem becomes significantly more challenging when multiple pedestrians are present in the occlusion.

The European Commission suggested a benchmark of 10^{-7} safety-critical incidents per hour of operation for functional and operational safety of autonomous vehicles [3]. Real-time vehicle testing is costly, time-consuming, and, in some cases, endangers the public. Therefore, it is important to investigate alternative methods for validation, such as simulations, formal proofs, fault injection, scenario-based testing, and human review [4]. Among these, scenario-based testing plays a key role in the safety validation of a CAV [5]. A scenario is defined as a pre-defined temporal sequence of events and actions that describes the development between several scenes in a sequence of scenes [6], [7]. As we integrate the shared situational awareness framework into the CAV system, it is crucial to evaluate and assess its effectiveness. To evaluate this integration, we propose to utilize a scenario-based test setup.

When multiple pedestrians are present within the ego-vehicle's occluded region, V2X units provide complementary viewpoints, enabling perception from different perspectives, as illustrated in Fig. 1. For instance, RSU 1 detects only one pedestrian, while RSU 2 detects both pedestrians, as shown by overlapping polygons. This introduces a significant challenge in associating measurements from multiple V2X units for multiple pedestrians. Determining which measurement corresponds to which pedestrian becomes difficult due to sensor noise and varying uncertainties. It is usually the case that each V2X unit maintains its own system for assigning IDs to detected pedestrians, leading to inconsistencies when data from multiple V2X units are combined. Moreover, pedestrians may also occlude one another from the line of sight of V2X units, making it challenging to associate and fuse heterogeneous perception information into a consistent representation.

In this work, we extend single pedestrian shared situational awareness framework [2] to track multiple occluded pedestrians. We develop a robust set-based estimation (SBE) algorithm for tracking, fusing, and predicting pedestrian motion. To ensure continuous tracking and measurement consistency across multiple V2X units, our algorithm links new measurements to existing tracks and employs cost-based association

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[1] Division of Decision and Control Systems, KTH Royal Institute of Technology, Stockholm, Sweden. {davnas, narri, kallej}@kth.se

[2] Research and Development, Scania CV AB, 151 87 Södertälje, Sweden. vandana.narri@scania.com

techniques to match tracks across different modules. To accurately handle the bounded uncertainty in measurements, we use a vertex-based representation of estimated position sets for pedestrians, which are then fused into a single, tighter, and guaranteed enclosure for each pedestrian's true location.

A. Related Work

Collaborative perception addresses two key limitations of on-board CAV sensors: occlusions and limited field of view. By exchanging perception information among V2X units, collaborative perception extends situational awareness beyond direct line-of-sight [8]. A detailed survey of various approaches for collaborative perception using V2X communication, datasets, and the challenges of deploying such methods in traffic scenarios is presented in [9]. This survey also highlights that real-world issues, such as communication imperfections, localization errors, and model discrepancies, can deteriorate system safety.

For multi-object tracking, [1] and [10] combine local single-object trackers with multi-hypothesis tracking to handle data association ambiguities. Subsequent works [11]–[14] employ cost-based association techniques for multi-object tracking in dense urban scenarios. To fuse outputs from multiple Kalman filters, covariance intersection and its variants are standard methods to ensure conservative consistency despite unknown cross-correlations [10]–[12].

On the other hand, set-based estimation offers a robust alternative to point-based Kalman filters by computing enclosures that provably contain the true position of the object under bounded uncertainty. To guarantee containment of true position of the object, set-based methods employ process models to propagate the prior estimated set and correct it by intersecting it with the measurement set at each time step [15], [16]. Zonotopes are popular set representations in set-based estimation due to their efficiency, but they are not suitable for exact intersections [16]–[18]. Alternative representations, such as constrained zonotopes (CZs) and V-polytopes, vary in performance depending on the system dimension. V-polytopes, while exact, become computationally infeasible in higher dimensions because the number of vertices grows exponentially [19]. However, in 2D, this bottleneck is minimal as polygons can be represented by a few vertices, making set operations manageable [20], [21]. V-polytopes also provide precise geometric representations, avoiding overapproximation caused by order reduction in CZs. For multi-pedestrian tracking in 2D, V-polytopes' accuracy and simplicity outweigh the efficiency benefits of CZs or zonotopes.

The reachable set for a given pedestrian is the set of all states that a pedestrian may reach over a given time horizon, subject to bounded velocity and acceleration [22]. Characterizing the pedestrian reachable set is fundamental for safe trajectory planning and control of CAVs in urban environments. In [23], pedestrian reachable sets are used to compute optimal vehicle maneuvers. A set-based prediction framework tailored to urban traffic rules is introduced in [24], demonstrating tight over-approximations of the pedestrian reachable set. More recently, [25] leverages data-driven mode segmentation to reduce conservativeness while preserving safety guarantees.

While probabilistic fusion techniques are well-established for Kalman filters, applying robust association and fusion logic to geometric, set-based estimates from multiple sources presents a distinct challenge, especially for multi-pedestrian scenarios. Moreover, accurately computing a reachable set without excessive over-approximation is crucial, as overly conservative estimated sets can lead to unnecessary restrictions in motion planning and control of CAVs. This need to avoid over-conservatism applies not only to the final reachable set for planning, but also to the core estimation loop itself, where incorporating multiple motion constraints can help systematically prune the estimated set at each time step. Finally, achieving computational efficiency in estimating reachable sets is essential to ensure real-time applicability in safety-critical situations such as multi-pedestrian tracking in occluded scenarios.

B. Contributions

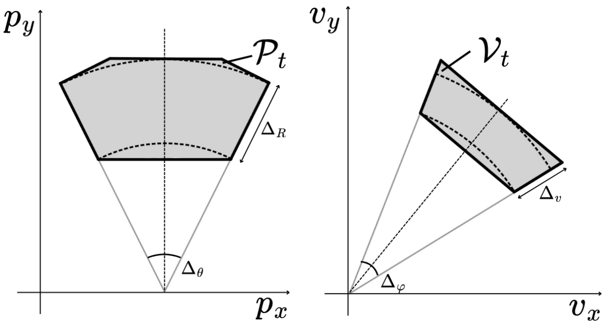
This paper introduces a novel set-based tracking framework for a CAV. The framework utilizes V2X communication and is tailored for multi-pedestrian scenarios in urban environments. Our contribution lies in establishing safety guarantees by systematically incorporating bounded uncertainties with unknown statistics, ensuring that the true pedestrian positions are robustly included in the estimated sets. To address the over-conservatism common in set-based multi-pedestrian tracking, we refine estimates by intersecting sensor data with pedestrian motion predictions generated from both bounded-velocity and acceleration constraints. Additionally, we implement a cost-based data association technique to accurately determine whether measurements from different V2X units pertain to the same or different pedestrians, enabling reliable data fusion across the network. The fusion method further improves these estimates, resulting in tighter sets for each pedestrian. This enables the prediction of future reachable sets, offering a reliable foundation for an ego-vehicle to plan a safe trajectory. Finally, the framework is validated through simulation on pedestrian trajectories from the UCY/Zara dataset [26], demonstrating its ability to produce tight, guaranteed bounds that outperform standard filtering methods.

C. Outline

The remainder of this paper is organized as follows. Section II describes the considered scenario (Fig. 1) of the multi-pedestrian tracking problem. Section III presents the proposed framework architecture and algorithm. Section IV provides simulation results and comparisons against existing methods. Finally, Section V provides conclusions and outlines future work.

II. Problem Formulation

In this section, we describe the considered scenario, define sensor measurements, provide preliminaries, and present the problem statement.



(a) Position set $\mathcal{P}_t$ (b) Velocity set $\mathcal{V}_t$

Fig. 2. Illustration of a convex measurement set from the RSUs. The dashed outlines indicate the original annular uncertainty sectors, while the solid gray regions show their convex overapproximations. (a) The position set $\mathcal{P}_t$ incorporates the uncertainty in bearing Δ_θ and range Δ_R; (b) the velocity set $\mathcal{V}_t$ incorporates the uncertainty in heading Δ_φ and speed Δ_v.

A. Scenario Description

Consider an occluded pedestrian-crossing scenario illustrated in Fig. 1. The setup consists of an ego-vehicle, a parked vehicle, two sensors (RSU1 and RSU2), and two pedestrians. The ego-vehicle is moving from left to right, and its field of view is shown as a blue sector. The parked vehicle creates an occlusion in the ego-vehicle's field of view, highlighted by a red sector. The field of views of RSU 1 and RSU 2 are represented by green and yellow sectors. Two pedestrians are located in the occluded region and are about to cross the road. Both RSUs can observe the occluded area and detect the two pedestrians. The measurements from the RSUs are then shared with the ego-vehicle.

B. Sensor Measurements

Each sensor provides a measurement set of positions and velocities, as illustrated in Fig. 2. The uncertainty in the position set originates from sensor errors in range and bearing, which is typically the case for cameras and LiDARs. At every time t, each sensor provides a measurement

$$M_t^d = \{(\mathcal{P}_t^{d,i}, \mathcal{V}_t^{d,i}, \mathrm{ID}^{d,i})\}_{i=0}^{m_t^d}$$

which comprises a collection of position sets $\mathcal{P}_t^{d,i}$, velocity sets $\mathcal{V}_t^{d,i}$, and identification numbers $\mathrm{ID}^{d,i}$, where $d = 1, \dots, D$ denotes the sensor index with D the total number of sensors, and i denotes the measurement index with m_t^d the total number of measurements sent by sensor d.

C. Vertex-Based Representation and Notation

In this paper, sets are represented by V-polytopes, which are defined as the convex hull of a finite number of vertices in $\mathbb{R}^2$, [27], [28]. Given the set of vertices $\mathbb{V} = \{v^{(1)}, \dots, v^{(r)}\}$, a V-polytope is given by $\mathcal{P} = \mathrm{conv}(\mathbb{V})$. By $\mathcal{D}_r \subset \mathbb{R}^2$, we denote a V-polytope with $r \geq 3$ vertices that circumscribes the unit disk $\{p \in \mathbb{R}^2 \mid \|p\|_2 \leq 1\}$.

The Minkowski sum of two V-polytopes $\mathcal{P}_1 = \mathrm{conv}(\mathbb{V}_1)$ and $\mathcal{P}_2 = \mathrm{conv}(\mathbb{V}_2)$, denoted $\mathcal{P}_1 \oplus \mathcal{P}_2$, is also a V-polytope given by

$$\mathcal{P}_1 \oplus \mathcal{P}_2 = \mathrm{conv}(\{v_1 + v_2 \mid v_1 \in \mathbb{V}_1,\ v_2 \in \mathbb{V}_2\}).$$

That is, the resulting polytope is the convex hull of all $|\mathbb{V}_1| \times |\mathbb{V}_2|$ pairwise sums of the vertices from $\mathcal{P}_1$ and $\mathcal{P}_2$.

The intersection $\mathcal{P}_1 \cap \mathcal{P}_2 = \{p \mid p \in \mathcal{P}_1 \text{ and } p \in \mathcal{P}_2\}$ of two V-polytopes is also a V-polytope (which may be empty). However, computing the vertex set of the resulting intersection polytope is a non-trivial problem, often requiring a conversion to a half-space representation or the use of more complex geometric algorithms (see, e.g., [29]).

The area of $\mathcal{P} = \mathrm{conv}(\mathbb{V})$ with vertices $v^{(i)} = (x^{(i)}, y^{(i)})$, for $i = 1, \dots, r$, is given by the Gauss'/shoelace formula

$$\begin{aligned} \mathrm{Area}(\mathcal{P}) = \frac{1}{2}\big| & x^{(1)}y^{(2)} + x^{(2)}y^{(3)} + \dots + x^{(r)}y^{(1)} \\ & - x^{(2)}y^{(1)} - x^{(3)}y^{(2)} - \dots - x^{(1)}y^{(r)}\big|. \end{aligned}$$

Finally, the intersection-over-union (IoU) metric is given by

$$\mathrm{IoU}(\mathcal{P}_1, \mathcal{P}_2) = \frac{\mathrm{Area}(\mathcal{P}_1 \cap \mathcal{P}_2)}{\mathrm{Area}(\mathcal{P}_1 \cup \mathcal{P}_2)}, \tag{1}$$

where $\mathrm{Area}(\mathcal{P}_1 \cup \mathcal{P}_2) = \mathrm{Area}(\mathcal{P}_1) + \mathrm{Area}(\mathcal{P}_2) - \mathrm{Area}(\mathcal{P}_1 \cap \mathcal{P}_2)$.

D. Problem Statement

In the scenario considered in Fig. 1, the ego-vehicle is soon approaching the crossing but does not have direct line-of-sight to the pedestrians. The ego-vehicle receives sensor measurements from RSU 1 and RSU 2. The GPS position of the originating RSU, along with its perception, is shared with the ego-vehicle in accordance with the collaborative perception message (CPM) service standard [30]. For the ego-vehicle to plan and execute a safe maneuver, it must be able to detect the number of pedestrians, estimate their current positions, and predict their reachable sets. If not, it might result in an unsafe situation for the pedestrian. With this objective in mind, we divide the problem statement into the following questions:

1) How to perform associations between multiple measurements and pedestrians?
2) How to estimate a robust set for each pedestrian's current position under varying levels of uncertainty across multiple V2X units?
3) How to compute reachable sets for each pedestrian without being overly conservative?

III. Methodology

This section describes the architecture and algorithm of the proposed framework.

A. Architecture

We propose the modular framework illustrated in Fig. 3 to manage and fuse data from multiple V2X units in a scalable manner. Each V2X unit, denoted by sensor d, produces a collection of measurements M_t^d at time t. These measurements are forwarded to the corresponding *estimation* module, where a track is created or updated for each pedestrian within the sensor's field of view. The resulting collection of tracks, denoted by T_t^d, is subsequently sent to the *track-to-track association* module. Each pedestrian may have tracks from multiple sensors that are then grouped in the track-to-track association module. This module employs a cost-based association algorithm to group tracks belonging to the same pedestrians. Each group is represented by a cluster C_t^p, where $p = 0, 1, \dots, P$ with P the total number of pedestrians in the scenario. Using these clusters, fused position set $\overline{\mathcal{P}}_t^p$ and velocity set $\overline{\mathcal{V}}_t^p$ are computed in the *fusion* module. The fused sets are passed on to the *global prediction* module, where reachable sets $\mathcal{R}_{t+T}^p$ for each pedestrian are computed.

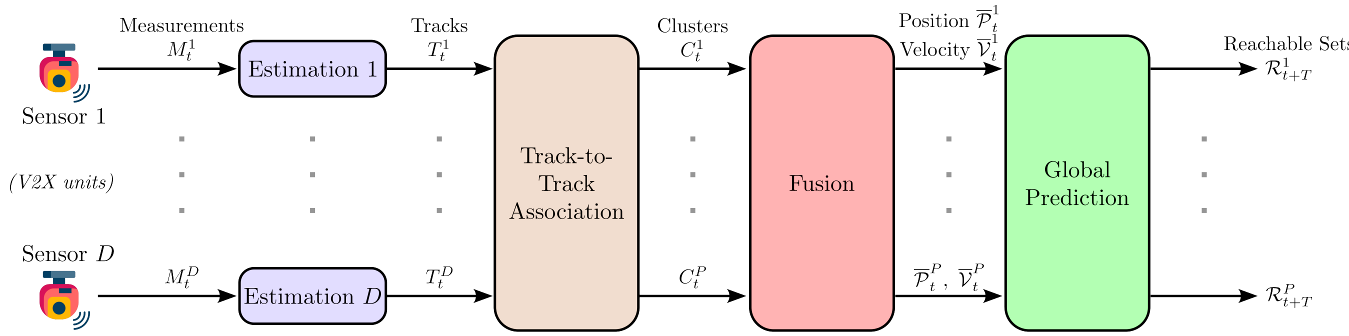


Fig. 3. Architecture of proposed framework. Reachable sets $\mathcal{R}^p_{t+T}$, $p = 1, \ldots, P$ for P pedestrians are computed using measurements from D sensors.

These reachable sets can be used by the ego-vehicle's motion planner to compute a safe trajectory.

B. Estimation

Each sensor independently tracks the pedestrians within its own field of view. This requires a dedicated estimation filter that can manage multiple pedestrian tracks, predict their motion, and correct these predictions using new sensor measurements. We define a *track* T_t^d as the collection of states maintained by the estimator d with different IDs. At time t, this collection is given by

$$T_t^d = \{(\widehat{\mathcal{P}}^{d,j}_{t|t}, \mathcal{V}^{d,j}_t, \widehat{\mathrm{ID}}^{d,j})\}_{j=1}^{n_t^d}$$

where j denotes the index for n_t^d active tracks. For each track j, $\widehat{\mathcal{P}}^{d,j}_{t|t}$ is the estimated position set, $\mathcal{V}^{d,j}_t$ is the velocity set, and $\widehat{\mathrm{ID}}^{d,j}$ is its persistent local identifier.

The estimation module follows a standard predict-correct cycle for each track j using the set of measurements M_t^d and the previous track set T^d_{t-1}.

1) Prediction Step

At each time step t, we predict the position set of each track j based on its previous estimated set $\widehat{\mathcal{P}}^{d,j}_{t-1|t-1}$. This prediction is based on two pedestrian motion constraints:

- *Bounded Velocity Prediction:* Assuming the pedestrian's bounded velocity[1] by $v^j_{\max} > 0$, the predicted set, denoted by $\check{\mathcal{P}}^{d,j}_{t|t-1}$, is the Minkowski sum of the previous position set and the set of admissible displacements under bounded velocity,
$$\check{\mathcal{P}}^{d,j}_{t|t-1} = \widehat{\mathcal{P}}^{d,j}_{t-1|t-1} \oplus \Delta t \cdot v^j_{\max} \mathcal{D}_r \tag{2}$$
where $\mathcal{D}_r$ is the polygon with r vertices circumscribing the unit disk and Δt is the sampling time.
- *Bounded Acceleration Prediction:* Employing the last known velocity $\mathcal{V}^{d,j}_{t-1}$ and assuming pedestrian's bounded acceleration[1] by $a^j_{\max} > 0$, the predicted set, denoted by $\tilde{\mathcal{P}}^{d,j}_{t|t-1}$, is computed as
$$\tilde{\mathcal{P}}^{d,j}_{t|t-1} = \widehat{\mathcal{P}}^{d,j}_{t-1|t-1} \oplus \Delta t \cdot \mathcal{V}^{d,j}_{t-1} \oplus \frac{1}{2} \Delta t^2 \cdot a^j_{\max} \mathcal{D}_r. \tag{3}$$

2) Correction Step

After prediction, we correct the estimate using the new measurements M_t^d. A local measurement-to-track association is performed by matching an incoming measurement's $\mathrm{ID}^{d,i}$ with the track's $\widehat{\mathrm{ID}}^{d,j}$. If a matching measurement i is found, the new estimate $\widehat{\mathcal{P}}^{d,j}_{t|t}$ (the *posteriori* estimate) is the intersection of the two predicted sets and the new measurement set $\mathcal{P}^{d,i}_t$. If no match is found, the track is propagated using only the more conservative bounded velocity prediction. This prevents the track from vanishing due to a missed detection at time t. The corrected estimate is computed as

$$\widehat{\mathcal{P}}^{d,j}_{t|t} = \begin{cases} \check{\mathcal{P}}^{d,j}_{t|t-1} \cap \tilde{\mathcal{P}}^{d,j}_{t|t-1} \cap \mathcal{P}^{d,i}_t, & \text{if } \exists i : \widehat{\mathrm{ID}}^{d,j} = \mathrm{ID}^{d,i} \\ \check{\mathcal{P}}^{d,j}_{t|t-1}, & \text{otherwise} \end{cases} \tag{4}$$

where $\check{\mathcal{P}}^{d,j}_{t|t-1}$ and $\tilde{\mathcal{P}}^{d,j}_{t|t-1}$ are given in (2) and (3), respectively. Similarly, the track's velocity set is also updated with the new measurement as

$$\mathcal{V}^{d,j}_t = \begin{cases} \mathcal{V}^{d,i}_t, & \text{if } \exists i : \widehat{\mathrm{ID}}^{d,j} = \mathrm{ID}^{d,i} \\ \emptyset, & \text{otherwise.} \end{cases} \tag{5}$$

The output of the estimation module is the updated set of tracks T_t^d for sensor d. Algorithm 1 summarizes the steps of an estimation module.

C. Track-to-Track Association

After performing the estimation steps, we have D independent sets of tracks $\{T_t^1, \ldots, T_t^D\}$ corresponding to D sensors. The track-to-track association module determines which position sets in the given set of tracks correspond to the same pedestrian. In this module, we solve a series of pairwise assignment problems. We compute a cost matrix $J^{(d_1,d_2)}$ for each pair of sensors (d_1, d_2), $d_1 \neq d_2$, where the cost of associating track j_1 from sensor d_1 with track j_2 from sensor d_2 is

$$J^{(d_1,d_2)}_{j_1 j_2} = \begin{cases} \ell^{(d_1,d_2)}_{j_1,j_2}, & \text{if } \widehat{\mathcal{P}}^{d_1,j_1}_{t|t} \cap \widehat{\mathcal{P}}^{d_2,j_2}_{t|t} \neq \emptyset \\ \infty, & \text{otherwise} \end{cases} \tag{6}$$

where
$\ell^{(d_1,d_2)}_{j_1,j_2} = \alpha(1 - \mathrm{IoU}(\widehat{\mathcal{P}}^{d_1,j_1}_{t|t}, \widehat{\mathcal{P}}^{d_2,j_2}_{t|t})) + \beta \|\widehat{c}^{d_1,j_1}_t - \widehat{c}^{d_2,j_2}_t\|_2$
with α and β the weighting coefficients, $\widehat{c}^{d_k,j_k}_t$ the centroid of the corresponding estimated position set $\widehat{\mathcal{P}}^{d_k,j_k}_{t|t}$ given in (4), $\mathrm{IoU}(\cdot)$ the Intersection-over-Union metric given in (1). In practice, when the intersection $\widehat{\mathcal{P}}^{d_1,j_1}_{t|t} \cap \widehat{\mathcal{P}}^{d_2,j_2}_{t|t}$ is empty, the cost is set to a large numeric value (e.g., 10^6) instead of ∞ to prevent association.

The optimal set of assignments for each pair is found using the Hungarian algorithm [31], a classic optimization algorithm for assignment problems. This algorithm takes the cost matrix $J^{(d_1,d_2)}$ in (6) and efficiently finds the optimal set of one-to-one pairings between the two sets of tracks that minimizes the total cost. In other words, this method

[1] $v^j_{\max}, a^j_{\max}$ can be selected based on the pedestrian's classification.

Algorithm 1 Estimation module

Input: $\Delta t, v_{\max}, a_{\max}$; $M_t^d = \{(\mathcal{P}_t^{d,i}, \mathcal{V}_t^{d,i}, \mathrm{ID}^{d,i})\}_{i=1}^{m_t^d}$; and $T_{t-1}^d = \{(\widehat{\mathcal{P}}_{t-1|t-1}^{d,j}, \mathcal{V}_{t-1}^{d,j}, \widehat{\mathrm{ID}}^{d,j})\}_{j=1}^{n_{t-1}^d}$

Output: $T_t^d = \{(\widehat{\mathcal{P}}_{t|t}^{d,k}, \mathcal{V}_t^{d,k}, \widehat{\mathrm{ID}}^{d,k})\}_{k=1}^{n_t^d}$ ▷ Updated tracks

$T_t^d \leftarrow \emptyset$ ▷ Initializing track collection
$U_t \leftarrow \{1, \dots, m_t^d\}$ ▷ Indices for unmatched IDs
for $j \leftarrow 1$ to n_{t-1}^d **do** ▷ Update existing tracks
  Compute $\check{\mathcal{P}}_{t|t-1}^{d,j}$ and $\tilde{\mathcal{P}}_{t|t-1}^{d,j}$ using (2) and (3)
  $found\_match \leftarrow 0$
  for $i \in U_t$ **do** ▷ Iterate to find a matching ID
    if $\mathrm{ID}^{d,i} == \widehat{\mathrm{ID}}^{d,j}$ **then**
      $\widehat{\mathcal{P}}_{t|t}^{d,j} \leftarrow \check{\mathcal{P}}_{t|t-1}^{d,j} \cap \tilde{\mathcal{P}}_{t|t-1}^{d,j} \cap \mathcal{P}_t^{d,i}$
      $\mathcal{V}_t^{d,j} \leftarrow \mathcal{V}_t^{d,i}$
      $U_t \leftarrow U_t \setminus \{i\}$ ▷ Remove measurement
      $found\_match \leftarrow 1$
      break ▷ Exit inner loop once ID matched
    end if
  end for
  if $found\_match == 0$ **then**
    $\widehat{\mathcal{P}}_{t|t}^{d,j} \leftarrow \check{\mathcal{P}}_{t|t-1}^{d,j}$
    $\mathcal{V}_t^{d,j} \leftarrow \emptyset$
  end if
  $T_t^d \leftarrow T_t^d \cup \{(\widehat{\mathcal{P}}_{t|t}^{d,j}, \mathcal{V}_t^{d,j}, \widehat{ID}^{d,j})\}$
end for
for $i \in U_t$ **do** ▷ Initiate new tracks for unmatched IDs
  $j \leftarrow j + 1$
  $(\widehat{\mathcal{P}}_{t|t}^{d,j}, \mathcal{V}_t^{d,j}, \widehat{\mathrm{ID}}^{d,j}) \leftarrow (\mathcal{P}_t^{d,i}, \mathcal{V}_t^{d,i}, \mathrm{ID}^{d,i})$
  $T_t^d \leftarrow T_t^d \cup \{(\widehat{\mathcal{P}}_{t|t}^{d,j}, \mathcal{V}_t^{d,j}, \widehat{\mathrm{ID}}^{d,j})\}$
end for
return T_t^d

finds the most likely set of correct matches (e.g., "track $j_1 = 1$ from RSU1 *is* track $j_2 = 3$ from RSU2") by finding the overall best fit that minimizes the total dissimilarity represented by $J^{(d_1,d_2)}$ across all pairings (d_1, d_2). These pairwise assignments are then chained together to form global clusters C_t^p. Each cluster C_t^p represents a single pedestrian p and contains at most one track from each sensor.

The complexity of the track-to-track association module can be split into two parts. The first part belongs to the pairwise combinations of sensors and the computation of the corresponding cost matrices for tracks. There are $\binom{D}{2} = D(D-1)/2$ pairs (d_1, d_2) of sensors and the size of each matrix $J^{(d_1,d_2)}$ is $n_t^{d_1} \times n_t^{d_2}$. Therefore, the complexity of computing each cost matrix $J^{(d_1,d_2)}$ is $n_t^{d_1} n_t^{d_2}$ times the complexity of computing $\mathrm{IoU}(\widehat{\mathcal{P}}_{t|t}^{d_1,j_1}, \widehat{\mathcal{P}}_{t|t}^{d_2,j_2})$, which is linear in the number of vertices of $\widehat{\mathcal{P}}_{t|t}^{d_1,j_1}$ and $\widehat{\mathcal{P}}_{t|t}^{d_2,j_2}$. The second part of the complexity belongs to the Hungarian method, which is $\mathcal{O}(\min(n_t^{d_1}, n_t^{d_2})^2 \max(n_t^{d_1}, n_t^{d_2}))$ for the cost matrix $J^{(d_1,d_2)}$. Therefore, the overall complexity of the track-to-track association module is polynomial of degree at most 3 in the number of pedestrians and linear in the maximum number of vertices in the position sets of tracks.

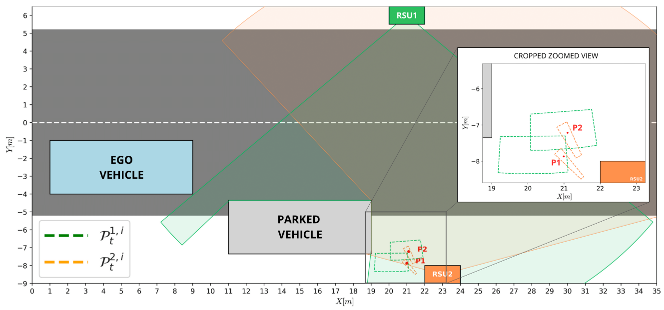


Fig. 4. Simulation scenario with two RSUs, shown in green and orange. The green and orange dashed-line polytopes indicate the measurement set from RSU 1 and RSU 2. The green and orange sectors represent the field of view of the respective RSUs. The red dots (•) denote the pedestrians' true positions.

D. Fusion

When the tracks are grouped into clusters C_t^p, we must fuse the information in each cluster to compute a single, unified, and more precise estimate for each pedestrian p. The fused position set $\overline{\mathcal{P}}_t^p$ is computed as the intersection of all estimated position sets $\widehat{\mathcal{P}}_{t|t}^{d,j}$ within the cluster C_t^p,

$$\overline{\mathcal{P}}_t^p = \bigcap_{T^{d,j} \in C_t^p} \widehat{\mathcal{P}}_{t|t}^{d,j} \tag{7}$$

This intersection operation leverages the different viewpoints of the sensors, as the overlapping area of their estimates provides a tighter bound on the pedestrian's true position.

Similarly, the fused velocity set $\overline{\mathcal{V}}_t^p$ is computed as the intersection of all non-empty velocity sets in (5) within the cluster C_t^p,

$$\overline{\mathcal{V}}_t^p = \bigcap_{T^{d,j} \in C_t^p : \mathcal{V}_t^{d,j} \neq \emptyset} \mathcal{V}_t^{d,j} \tag{8}$$

The output of this module is a single fused state $(\overline{\mathcal{P}}_t^p, \overline{\mathcal{V}}_t^p)$ for each tracked pedestrian p.

E. Global Predictions

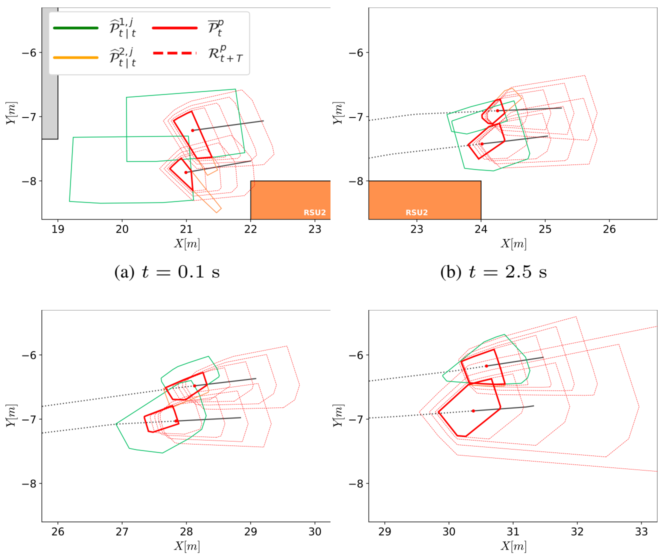
Finally, to provide a safe and reliable input to the ego-vehicle's motion planner, we predict the future positions of each tracked pedestrian. We compute the future reachable set $\mathcal{R}_{t+T}^p$ for each tracked pedestrian p over a prediction horizon $T = h\Delta t$ with h a positive integer. This prediction is based on the fused state $(\overline{\mathcal{P}}_t^p, \overline{\mathcal{V}}_t^p)$ and uses the same two motion constraints:

- *Bounded Velocity Reachable Set:*
$$\check{\mathcal{R}}_{t+T}^p = \overline{\mathcal{P}}_t^p \oplus T \cdot v_{\max}^p \mathcal{D}_r$$
- *Bounded Acceleration Reachable Set:*
$$\hat{\mathcal{R}}_{t+T}^p = \overline{\mathcal{P}}_t^p \oplus T \cdot \overline{\mathcal{V}}_t^p \oplus \frac{1}{2} T^2 \cdot a_{\max}^p \mathcal{D}_r$$

The final reachable set $\mathcal{R}_{t+T}^p$ for each tracked pedestrian p is given by

$$\mathcal{R}_{t+T}^p = \begin{cases} \check{\mathcal{R}}_{t+T}^p \cap \hat{\mathcal{R}}_{t+T}^p, & \text{if } \overline{\mathcal{V}}_t^p \neq \emptyset \\ \check{\mathcal{R}}_{t+T}^p, & \text{otherwise.} \end{cases} \tag{9}$$

This set $\mathcal{R}_{t+T}^p$ provides a guaranteed enclosure of the pedestrian's true trajectory during the time horizon $[t, t+T]$, which the ego-vehicle's motion planner can directly use to ensure safety.



(a) $t = 0.1$ s (b) $t = 2.5$ s (c) $t = 5.0$ s (d) $t = 7.5$ s

Fig. 5. Simulation results of the considered scenario at four time instances.

IV. Evaluation

This section evaluates the proposed framework's performance on the scenario illustrated in Fig. 1. This scenario depicts a complex setting in which pedestrians are in close proximity to one another, making it difficult to track and localize them.

A. Simulation Setup

The scenario in Fig. 4 is a Python-based simulation environment[2] created based on Fig. 1. In Fig. 4, the ego-vehicle (light blue) communicates with two RSUs (green and orange). The parked vehicle (gray) creates an occlusion in the ego-vehicle's field of view. The ground truth for two pedestrians is shown as red dots. These pedestrians' trajectories are from the UCY/Zara dataset [26]. Measurements are generated by introducing random noise to these trajectories, which are sampled at time intervals of $\Delta t = 0.1\,\mathrm{s}$. We consider the uncertainty bounds for range and bearing as $\Delta_R = \pm 0.5\,\mathrm{m}$ and $\Delta_\theta = \pm 4°$, which are from vision-based pedestrian tracking and localization models [32], [33]. We consider the maximum pedestrian velocity and acceleration as $2\ \mathrm{m/s}$ and $0.6\ \mathrm{m/s^2}$, which is based on [24].

In Fig. 4, we plot the measurement sets $\mathcal{P}_t^{1,i}$ and $\mathcal{P}_t^{2,i}$ for the positions of the pedestrians originating from RSU 1 and RSU 2. Note that the sets $\mathcal{P}_t^{1,i}$ are larger than the sets $\mathcal{P}_t^{2,i}$ because RSU 1 is farther from the pedestrians, causing the same angular bearing uncertainty Δ_θ to subtend a larger area.

B. Simulation Results

The results from the proposed framework are presented in Fig. 5 for four time instances. The estimated sets $\widehat{\mathcal{P}}_t^{1,j}$ and $\widehat{\mathcal{P}}_t^{2,j}$ computed using measurements from RSU 1 and RSU 2 are illustrated as green and orange solid-line polytopes. The fused sets $\overline{\mathcal{P}}_t^p$ are shown as red solid-line polytopes, and the global reachable sets $\mathcal{R}_{t+T}^p$, for $T \in \{0.2, 0.4, 0.6, 0.8\}$, are represented by red dotted-line polytopes. Black dotted lines depict the past trajectories and black solid lines depict the future trajectories of pedestrians.

At $t = 0.1\,\mathrm{s}$, the first set of measurements M_t^1 and M_t^2 for the two pedestrians are received from both RSUs. At this initial step, we initialize tracks T_t^1 from RSU 1 and T_t^2 from RSU 2. Consequently, the first loop from line 3 to 20 in Algorithm 1 is bypassed as $n_{t-1}^d = 0$. Instead, the second loop from line 21 to 25 in Algorithm 1 is executed. For new track, the first estimated set is same as the measurement set (i.e., $\widehat{\mathcal{P}}_{t|t}^{d,j} \leftarrow \mathcal{P}_t^{d,i}$). Therefore, the green and orange polytopes shown in Fig. 5a represent both the initial measurements and the initial estimated sets.

These tracks T_t^1 and T_t^2 are passed to the track-to-track association module. This module groups the given tracks into two clusters C_t^1 and C_t^2. The fusion module then computes the fused position sets $\overline{\mathcal{P}}_t^1$ and $\overline{\mathcal{P}}_t^2$. These sets are computed by intersecting the estimated sets within each cluster, as per (7). Finally, the global prediction module uses these fused position sets ($\overline{\mathcal{P}}_t^1$ and $\overline{\mathcal{P}}_t^2$) and the fused velocity sets ($\overline{\mathcal{V}}_t^1$ and $\overline{\mathcal{V}}_t^2$) to compute the future reachable sets $\mathcal{R}_{t+T}^1$ and $\mathcal{R}_{t+T}^2$.

Fig. 5b presents a slightly later instance, when both pedestrians are detected by the RSU 1 and RSU 2. The estimated and fused sets for these two pedestrians are smaller than in Fig. 5a, as are the global predictions. The cluster association cost matrix at $t = 2.5\,\mathrm{s}$ is

$$J^{(1,2)} = \begin{bmatrix} -0.504 & -0.010 \\ 10^6 & -0.077 \end{bmatrix}$$

where the rows correspond to the tracks from RSU 1 and the columns to those from RSU 2. The $(2,1)$ entry is set to 10^6 because the tracks are disjoint. Fig. 5c represents a time instance at $t = 5.0\,\mathrm{s}$, the continuous fusion of measurements results in tighter estimated sets $\widehat{\mathcal{P}}_{t|t}^{d,j}$ and fused sets $\overline{\mathcal{P}}_t^p$ compared to Fig. 5a.

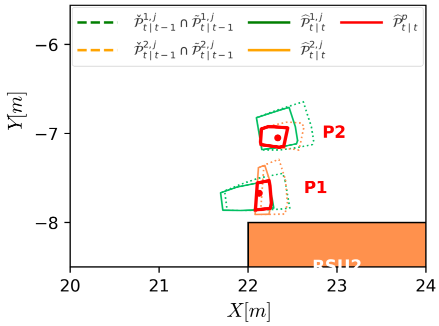
Fig. 5d at $t = 7.5\,\mathrm{s}$ illustrates a key event where RSU 2 has not been detecting P1 for the last 2 seconds, continuously triggering the `if(found_match==0)` condition in line 15 of Algorithm 1. As per the algorithm, the estimated set $\widehat{\mathcal{P}}_{t|t}^{2,1}$ has been propagating based on the bounded-velocity prediction $\check{\mathcal{P}}_{t|t-1}^{2,1}$, which causes the set to become conservatively large. Therefore, we remove the estimated set $\widehat{\mathcal{P}}_{t|t}^{2,1}$, the empty velocity set $\mathcal{V}_t^{2,1}$, and the $\widehat{\mathrm{ID}}^{2,1}$ corresponding to P1 from RSU 2 at $t = 7.5\,\mathrm{s}$.[3] This causes the fused set for P1 to be equal to the estimated set from RSU 1 (the green polytope). Since there are no other estimated sets for P1 to reduce the uncertainty of the fused set, the reachable sets are larger than in the previous time steps. The computational complexity of these operations depends linearly on the number of vertices, which remained consistently low (approximately less than 11) in our simulations. This low vertex count enables average simulation time to be under 20 μs.

C. Performance Comparison

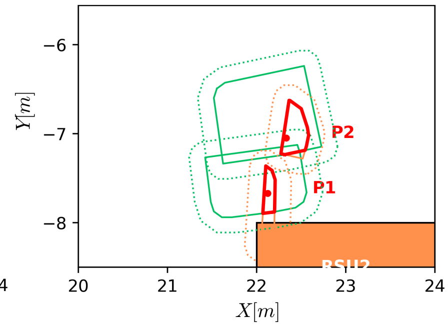
The performance of the proposed framework is compared with two estimation methods: an SBE method that uses only a

[2] https://github.com/davidenascivera/CuttingThroughUncertaintyV2X

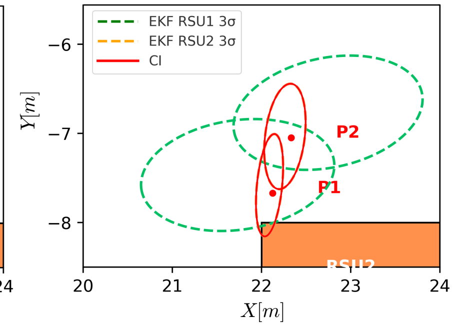
[3] Algorithm 1 does not specify a mechanism for removing tracks when a pedestrian is consistently undetected by an RSU. However, our code implements this feature.



(a) Proposed SBE method



(b) SBE with a maximum velocity



(c) EKF with covariance intersection

Fig. 6. Comparison of estimated and fused sets for a scenario with only two pedestrians, computed using (a) the proposed set-based estimation (SBE) incorporating maximum velocity and acceleration constraints, (b) SBE with only a maximum velocity constraint, and (c) extended Kalman filter (EKF) with covariance intersection. The red dots (•) represent the pedestrians' true position.

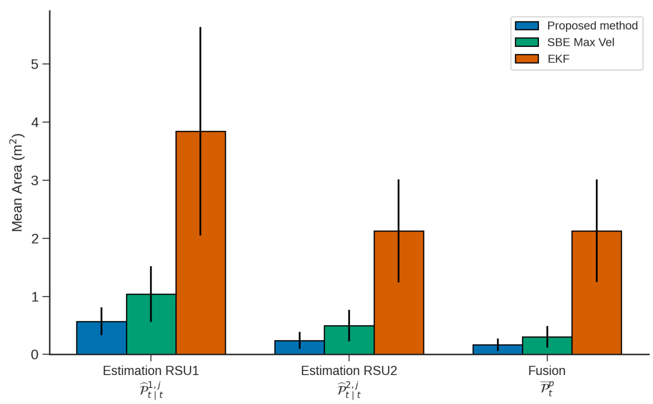


Fig. 7. Comparison of the mean area of the estimated and fused sets for different methods. The figure illustrates the over approximation of the resultant set by (a) the proposed method, (b) the SBE with only a maximum velocity constraint, and (c) the EKF with covariance intersection.

bounded-velocity model for reachability and an EKF method with covariance intersection. Here, covariance intersection is a conservative fusion technique that combines two Gaussian estimates using a weighted combination of their information matrices when cross-correlation is unknown. A comparison of these methods is presented in Fig. 6, which illustrates the estimated sets at $t = 0.90\,\mathrm{s}$. Note that the estimates from RSU 1 are RSU 2 are given by green and orange polygons. The fused sets are given by red polygons. The sets generated by the proposed method, which incorporates the maximum velocity and acceleration bounds, are shown in Fig. 6a and are tighter than those produced by the SBE with a maximum velocity bound, shown in Fig. 6b. The EKF with covariance intersection, shown in Fig. 6c, where the 3σ of Gaussian estimates (99.7% confidence) are considered as the estimated set and the fused set. This result confirms the proposed framework's ability to achieve less conservative estimate of the pedestrian location than alternative methods.

A quantitative analysis is provided in Fig. 7, which presents the mean and standard deviation of the computed set areas. The chart is divided into three groups: the individual estimates from RSU 1 and RSU 2, corresponding to their estimated sets $\widehat{\mathcal{P}}^{d,j}_{t|t}$, and the final fused set $\overline{\mathcal{P}}^{p}_{t}$. It can be observed that the proposed method consistently yields the smallest mean area, demonstrating less conservativeness throughout the simulation. The fused set obtained by the proposed method achieves a mean area of $0.169 \pm 0.106\,\mathrm{m}^2$. In contrast, the alternative methods yield significantly larger mean areas of the fused set: the SBE with only a maximum velocity bound results in $0.304 \pm 0.188\,\mathrm{m}^2$, while the EKF with covariance intersection results in a mean area of $2.129 \pm 0.884\,\mathrm{m}^2$. For context, the mean areas of the original measurement sets $\mathcal{P}^{d,i}_{t}$ are $2.589 \pm 0.197\,\mathrm{m}^2$ (from RSU1) and $0.978 \pm 0.405\,\mathrm{m}^2$ (from RSU2).

V. Conclusion

This work presented a robust, end-to-end framework for multi-pedestrian tracking using V2X communication and set-based estimation. The proposed framework provides guaranteed safety enclosures of pedestrian positions in occluded environments. There are two complementary processes that systematically reduce over-conservatism. First, the track association method correctly groups measurements from multiple sensors to enable fusion. Second, the set-based estimation loop prunes uncertainty by intersecting measurements with predictions under bounded velocity and acceleration constraints. This combination ensures the guaranteed sets remain tight and actionable for an ego-vehicle's planner, which is a significant improvement over standard filtering techniques that do not offer the same formal guarantees. We also demonstrated the real-time feasibility of the proposed framework through simulations, as the geometric operations required for 2D tracking (estimation, fusion, and reachability) are tractable. This confirms that the formally guaranteed reachable sets can be generated and provided to a motion planner in real-time.

Several promising avenues for future research are open. The most critical next step is to validate this framework on real-world V2X hardware, which will introduce new challenges such as network latency, packet loss, and sensor synchronization. Beyond validation, the track association method could be significantly enhanced. Instead of relying purely on geometric overlap, future work should incorporate contextual information from high-definition maps. This would not only improve association accuracy but also allow the reachability analysis to be constrained by environmental context (e.g., sidewalks, crosswalks, or physical barriers), drastically reducing the conservatism of the predicted sets. Finally, the generic motion constraints could be replaced with more sophisticated, data-driven predictors. A promising direction is the development of learning-based reachability methods that incorporate different pedestrian behaviors (e.g., "waiting to cross" vs. "running"), which would improve the accuracy and semantic utility of the predicted sets.

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